

Physics-0 2026 Spring

Homework 6

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*All of my homework is available at <https://github.com/lacampanella233/Homework.git>.

Problem 1. (20pts) Consider a spacecraft of mass m at rest on the surface of a planet of mass M and radius R . How much work must the spacecraft engines perform to move the spacecraft to a circular orbit a distance R from the surface of the planet?

Solution. When a spacecraft is on a circular orbit of radius $2R$, the gravitational force provides the centripetal force. Denote the speed of the spacecraft on the circular orbit as v . We have

$$m \frac{v^2}{2R} = \frac{GMm}{(2R)^2} \implies v = \sqrt{\frac{GM}{2R}}$$

Then the mechanical energy of the spacecraft on the circular orbit is

$$E = \frac{1}{2}mv^2 - \frac{GMm}{2R} = -\frac{GMm}{4R}$$

The mechanical energy of the spacecraft on the surface of the planet is

$$E_0 = -\frac{GMm}{R}$$

Therefore, the work that the spacecraft engines must perform to move the spacecraft to the circular orbit is

$$W = E - E_0 = \frac{3GMm}{4R}$$

Problem 2. (30pts) Consider two spheres with radius R and masses m and $2m$ in empty space (far away from any other forces). The spheres are released from rest with their centers a distance d apart. They accelerate towards each other due to their gravitational attraction.

- (a) (15pts) Compute the relative velocity between the spheres when they collide (note that the spheres have radii R and cannot be treated as point particles).
- (b) (15pts) How far does the first sphere (of mass m) move when the spheres collide?

Solution. Since the gravitational potential produced by a sphere outside itself is the same as that produced by a point mass located at its center, we can treat the two spheres as point masses located at their centers when the distance between their centers is greater than $2R$.

Denote the speed of the lighter sphere in the center-of-mass frame as v_1 and the speed of the heavier sphere as v_2 . Since the total momentum is conserved, we have

$$mv_1 = 2mv_2 \implies v_1 = 2v_2$$

The total mechanical energy of the system is conserved, then we have

$$-\frac{2Gm^2}{d} = \frac{1}{2}mv_1^2 + \frac{1}{2}(2m)v_2^2 - \frac{2Gm^2}{2R} \implies v_1 = 2v_2 = 2\sqrt{\frac{Gm(d-2R)}{3dR}}.$$

Then the relative velocity between the spheres when they collide is

$$v_1 + v_2 = \sqrt{\frac{3Gm(d-2R)}{dR}}.$$

(b) Denote the displacement of the lighter sphere in the center-of-mass frame as x_1 and the displacement of the heavier sphere as x_2 . Since the total momentum is conserved, we have

$$mx_1 = 2mx_2, \quad x_1 + x_2 = d - 2R \implies x_1 = \frac{2}{3}(d - 2R).$$

Problem 3. (20pts) The density of planets is generally not uniform: they are densest at the center and have decreasing density toward the surface. Model a planet as a spherically symmetric object with the same radius as the Earth, and a density that decreases linearly with distance from the center. Let the densities at the center and the surface of the planet be $15 \times 10^3 \text{ kg/m}^3$ and $2 \times 10^3 \text{ kg/m}^3$, respectively. Find the acceleration due to gravity at the surface of this planet.

Solution. Denote the radius of Earth as $R_0 = 6.4 \times 10^6 \text{ m}$ and the density at the center and the surface of the planet as $\rho_c = 15 \times 10^3 \text{ kg/m}^3$ and $\rho_s = 2 \times 10^3 \text{ kg/m}^3$, respectively. Since the density decreases linearly with distance from the center, we can express the density as a function of radius r as

$$\rho(r) = \rho_c - \left(\frac{\rho_c - \rho_s}{R_0} \right) r.$$

The total mass of the planet can be calculated by integrating the density over the volume of the planet:

$$M = \int_0^{R_0} 4\pi r^2 \rho(r) dr = \int_0^{R_0} 4\pi r^2 \left(\rho_c - \left(\frac{\rho_c - \rho_s}{R_0} \right) r \right) dr = \pi R_0^3 \left(\frac{\rho_c}{3} + \rho_s \right).$$

The acceleration due to gravity at the surface of the planet is

$$a = \frac{GM}{R_0^2} = \pi G R_0 \left(\frac{\rho_c}{3} + \rho_s \right).$$

Substituting the values of G , R_0 , ρ_c , and ρ_s gives

$$a = 9.35 \text{ m/s}^2.$$

Problem 4. (30pts) Let $f(x, y, z) = x^2y^2 + y^2z^2 + z^2x^2$.

(a) (10pts) Compute the vector field $\mathbf{F} = \nabla f$. Sketch the vector field on the xy -plane with $z = 0$.

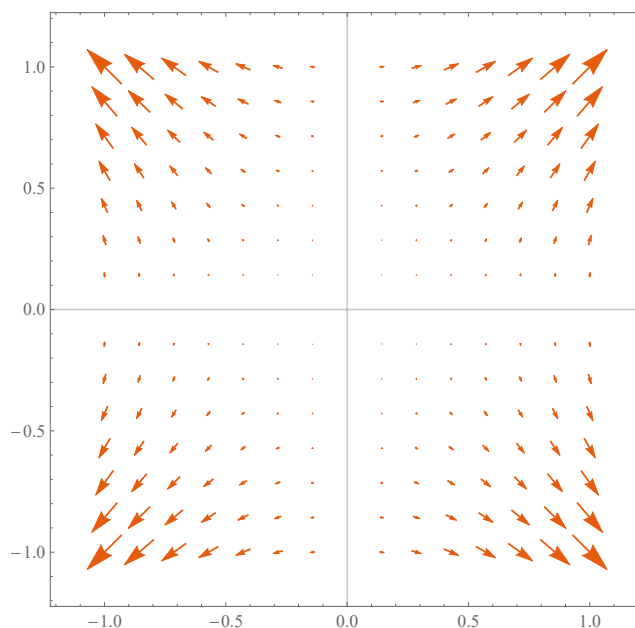
(b) (10pts) Compute the divergence and the curl of $\mathbf{F}$.

(c) (10pt) Consider an arbitrary scalar field $\phi(x, y, z)$. Assuming that Φ is a C^2 function (such that all of its first and second partial derivatives exist and are continuous), show that all of the curl of the gradient of ϕ vanishes.

Solution. (a) We have

$$\mathbf{F} = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = (2xy^2 + 2xz^2, 2x^2y + 2yz^2, 2y^2z + 2x^2z).$$

The sketch of the vector field on the xy -plane with $z = 0$ is shown below:



(b) The divergence of $\mathbf{F}$ is

$$\begin{aligned} \nabla \cdot \mathbf{F} &= \frac{\partial}{\partial x}(2xy^2 + 2xz^2) + \frac{\partial}{\partial y}(2x^2y + 2yz^2) + \frac{\partial}{\partial z}(2y^2z + 2x^2z) \\ &= 2y^2 + 2z^2 + 2x^2 + 2z^2 + 2y^2 + 2x^2 \\ &= 4x^2 + 4y^2 + 4z^2. \end{aligned}$$

The curl of $\mathbf{F}$ is 0 according to (c).

(c) We have

$$\begin{aligned} \nabla \times (\nabla \phi) &= (\partial_y(\nabla \phi)_z - \partial_z(\nabla \phi)_y, \partial_z(\nabla \phi)_x - \partial_x(\nabla \phi)_z, \partial_x(\nabla \phi)_y - \partial_y(\nabla \phi)_x) \\ &= (\partial_y \partial_z \phi - \partial_z \partial_y \phi, \partial_z \partial_x \phi - \partial_x \partial_z \phi, \partial_x \partial_y \phi - \partial_y \partial_x \phi) \\ &= (0, 0, 0). \end{aligned}$$

Note that ϕ is a C^2 function, so the order of partial derivatives can be interchanged.